Extended abstract (2026)

Left-to-right minima basis compatible with the descent-algebra filtration

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Abstract. We introduce a new basis of the group algebra of the symmetric group, built using the left-to-right minima sets of permutations. We show that on this basis, the descent algebra acts by triangular operators, thus making it an analogue of a cellular basis. The proof involves Dynkin elements (nested commutators) of the free algebra and their interactions with the **B**-basis.

Résumé. Nous introduisons une nouvelle base de l'algèbre de groupe du groupe symétrique, construite à partir des ensembles de minima gauches des permutations. Nous montrons que, sur cette base, l'algèbre de descente agit par opérateurs triangulaires, ce qui en fait un analogue d'une base cellulaire. La démonstration fait intervenir les éléments de Dynkin (commutateurs imbriqués) de l'algèbre libre et leurs interactions avec la base **B**.

Keywords: Left-to-right minima, group algebra, descent algebra, Dynkin operator

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- Full native symmetric-group/rook-sum/left-minima/shuffle algebra framework. All original coefficient-ring, factorial-invertibility, composition/refinement and index restrictions retained with their exact definitions.. Complete author source-local straightening, Dynkin/permutation identities and annihilator arguments supplied; no external proof reconstruction or independent matrix/permutation calculation.

1 Introduction

For a nonnegative integer n, let S_n denote the symmetric group on $[n] = \{1, 2, \ldots, n\}$. Fix a ground ring $\mathbf{k}$ and let $\mathcal{A} = \mathbf{k}[S_n]$ be the group algebra of S_n over $\mathbf{k}$. For $w \in S_n$, the *descent set* of w is

$$\mathrm{Des}(w) = \{\, i \in [n-1] \mid w(i) > w(i+1) \,\}.$$

For every subset $I \subseteq [n-1]$ define the element

$$\mathbf{B}_I = \sum_{\substack{w \in S_n; \\ \mathrm{Des}(w) \subseteq I}} w \in \mathcal{A}.$$

The span of the elements $\mathbf{B}_I$ is the classical *descent algebra* $\Sigma_n \subseteq \mathcal{A}$ introduced by Solomon in [8]. We call the family $(\mathbf{B}_I)_{I \subseteq [n-1]}$ the $\mathbf{B}$-*basis* of Σ_n. The aim of this paper is to construct and study a new basis $(\beta_w)_{w \in S_n}$ of the whole algebra $\mathcal{A}$ which is compatible with the right-action of the descent algebra in the following sense: there should exist a partial order $\preceq$ on S_n such that for every $w \in S_n$ and every $I \subseteq [n-1]$ one has

$$\mathbf{B}_I \cdot \beta_w \ \in \ \mathrm{span}\{\, \beta_u \mid u \preceq w \,\}.$$

In other words, the action of Σ_n on $\mathcal{A}$ should be represented by triangular matrices on this basis. Our basis will furthermore be compatible with a natural family of right ideals of $\mathcal{A}$ generated by the **B**-basis. In the sequel we introduce a poset-indexed filtration of $\mathcal{A}$ built from the right ideals $\mathbf{B}_\alpha\mathcal{A}$ (reindexing $\mathbf{B}_I$ by compositions), construct our candidate basis, and prove that it is a basis adapted to the filtration.

The paper is organised as follows. In Section 2 we define the filtration, recall the necessary combinatorics of integer compositions, and record basic properties of the **B**-basis action. Section 3 introduces the left-to-right-minima basis and states the main theorem connecting it with the filtration. The remaining sections are devoted to proofs and examples.

2 A poset-indexed filtration of the group algebra

2.1 Compositions and refinement

A *composition* $\alpha = (\alpha_1, \ldots, \alpha_p)$ of a nonnegative integer n (notation $\alpha \vDash n$) is a finite sequence of positive integers with $|\alpha| := \sum_i \alpha_i = n$; write $\ell(\alpha) = p$ for its length. Let $\widetilde{\alpha}$ denote the *underlying partition* of α, that is, the partition obtained by rearranging the parts of α in weakly decreasing order. Two compositions α and β will be called *anagrams* (of each other) if they satisfy $\widetilde{\alpha} = \widetilde{\beta}$. Let Comp_n be the set of compositions of n. A well-known bijection between compositions of n and subsets of $[n-1]$ sends each composition $\alpha = (\alpha_1, \ldots, \alpha_p) \in \mathrm{Comp}_n$ to

$$\mathrm{Set}(\alpha) := \{\alpha_1,\ \alpha_1 + \alpha_2,\ \ldots,\ \alpha_1 + \cdots + \alpha_{p-1}\} \subseteq [n-1],$$

and its inverse map sends each subset $I = \{i_1 < \cdots < i_{p-1}\} \subseteq [n-1]$ to $\mathrm{Comp}(I) = (i_1 - i_0, i_2 - i_1, \ldots, i_p - i_{p-1}) \in \mathrm{Comp}_n$, where we set $i_0 := 0$ and $i_p := n$. Accordingly we reindex the **B**-basis by compositions by setting $\mathbf{B}_\alpha := \mathbf{B}_{\mathrm{Set}(\alpha)}$ for $\alpha \in \mathrm{Comp}_n$.

Compositions are partially ordered by refinement:

Definition 1 (refinement)**.** *Let $\alpha = (\alpha_1, \ldots, \alpha_m)$ and $\beta = (\beta_1, \ldots, \beta_p)$ be compositions of the same integer n. We say that α* refines *β, written $\alpha \preceq \beta$, if α can be partitioned into p contiguous sub-compositions $\alpha^{(1)}, \ldots, \alpha^{(p)}$ with $|\alpha^{(j)}| = \beta_j$ for all j.*

Partitions are also partially ordered by refinement, but of a different kind:

Definition 2 (partition refinement)**.** *Let $\lambda, \mu \vdash n$ be partitions of n. We say that λ is a* partition refinement *of μ, written $\lambda \preceq_\pi \mu$, if some anagram of λ refines μ (as a composition).*

Equivalently, this relation $\preceq_\pi$ can be defined as follows: Two partitions $\lambda = (\lambda_1, \ldots, \lambda_k)$ and $\mu = (\mu_1, \ldots, \mu_l)$ satisfy $\lambda \preceq_\pi \mu$ if and only if there exists a map $f : [k] \to [l]$ such that each $j \in [l]$ satisfies $\mu_j = \sum \lambda_i$, where the sum ranges over all $i \in f^{-1}(j)$.

Both relations $\preceq$ and $\preceq_\pi$ are weak partial orders; their strict versions will be called $\prec$ and $\prec_\pi$. Note that $\alpha \preceq \beta$ implies $\widetilde{\alpha} \preceq_\pi \widetilde{\beta}$.

For any composition $\alpha = (\alpha_1, \ldots, \alpha_k) \vDash n$, we define the *$\alpha$-partial sums* to be the $k+1$ integers

$$\alpha_{\leq i} := \alpha_1 + \alpha_2 + \cdots + \alpha_i \qquad \text{for all } 0 \leq i \leq k,$$

and we define the *α-blocks* to be the k subsets[1]

$$\mathrm{Set}(\alpha)_i := [\alpha_{\leq i-1} + 1,\ \alpha_{\leq i}] \text{ of } [n] \qquad \text{for all } i \in [k]. \tag{2.1}$$

These α-blocks $\mathrm{Set}(\alpha)_1, \mathrm{Set}(\alpha)_2, \ldots, \mathrm{Set}(\alpha)_k$ are k disjoint nonempty integer intervals of respective lengths $\alpha_1, \alpha_2, \ldots, \alpha_k$, and their union is $[n]$. The $\mathbf{B}$-basis element $\mathbf{B}_\alpha$ is then the sum of all permutations $w \in S_n$ that increase on each α-block.

2.2 Action of the B-basis and right ideals

Elements of the $\mathbf{B}$-basis can be conveniently multiplied using the *Solomon's Mackey formula* (see e.g. [9, Theorem 2]). If M is a nonnegative integer matrix (i.e., matrix with nonnegative integer entries), denote by $\mathrm{rows}(M)$ and $\mathrm{cols}(M)$ its row- and column-sum vectors and by $\mathrm{read}(M)$ the composition obtained by concatenating the rows of M (omitting zeros). Given $\alpha, \beta \in \mathrm{Comp}_n$, write $\mathbb{N}^{\alpha,\beta}$ for the set of nonnegative integer matrices with column sums α and row sums β. Solomon's formula (in our notation) reads

$$\mathbf{B}_\alpha \mathbf{B}_\beta = \sum_{M \in \mathbb{N}^{\alpha,\beta}} \mathbf{B}_{\mathrm{read}(M)}. \tag{2.2}$$

In particular every composition $\mathrm{read}(M)$ on the right refines β and satisfies $\widetilde{\mathrm{read}(M)} \preceq_\pi \widetilde{\alpha}$; consequently left multiplication by $\mathbf{B}_\alpha$ is triangular on the basis $(\mathbf{B}_\gamma)_{\gamma \in \mathrm{Comp}_n}$ of Σ_n with respect to refinement.

Definition 3 ($\mathbf{B}$-right ideals)**.** *For $\beta \in \mathrm{Comp}_n$ define the right ideal*

$$\mathcal{R}_\beta := \mathbf{B}_\beta \mathcal{A} \subseteq \mathcal{A}.$$

It is easy to see that the right ideal $\mathcal{R}_\alpha$ depends only on the multiset of parts of α (i.e. on the corresponding integer partition), not on their order.

Lemma 1. *If $\alpha, \beta \in \mathrm{Comp}_n$ are anagrams (that is, $\widetilde{\alpha} = \widetilde{\beta}$), then $\mathcal{R}_\alpha = \mathcal{R}_\beta$.*

Proof. If β is an anagram of α, then there exists a permutation $w \in S_n$ sending the α-blocks to the β-blocks while preserving the order of the elements in each block. Thus, we easily find $\mathbf{B}_\beta w = \mathbf{B}_\alpha$, and therefore $\mathbf{B}_\alpha \mathcal{A} = \mathbf{B}_\beta w \mathcal{A} = \mathbf{B}_\beta \mathcal{A}$. □

The family $(\mathcal{R}_\beta)_{\beta \in \mathrm{Comp}_n}$ is a poset-indexed filtration of $\mathcal{A}$ (ordered by refinement).

[1] The notation $[u, v]$ for two integers u and v denotes the integer interval $\{x \in \mathbb{Z} \mid u \leq x \leq v\}$.

Lemma 2. *If $\beta, \gamma \in \mathrm{Comp}_n$ and $\beta \preceq \gamma$ (i.e. β refines γ), then $\mathcal{R}_\beta \subseteq \mathcal{R}_\gamma$.*

Proof. Write the composition γ as $\gamma = (\gamma_1, \gamma_2, \ldots, \gamma_k)$, and let

$$(I_1, I_2, \ldots, I_k) := (\mathrm{Set}\,(\gamma)_1, \mathrm{Set}\,(\gamma)_2, \ldots, \mathrm{Set}\,(\gamma)_k)$$

be the list of the γ-blocks. Let $\mathbf{B}_{\gamma\to\beta}$ be the sum of those permutations $w \in S_n$ that preserve the γ-blocks (i.e., satisfy $w(I_p) = I_p$ for all p) and satisfy $\mathrm{Des}\, w \subseteq \mathrm{Set}\,(\beta)$. By the definition of $\mathbf{B}_\beta$, we have

$$\mathbf{B}_\beta = \sum_{\substack{w\in S_n;\\ \mathrm{Des}\, w\subseteq \mathrm{Set}(\beta)}} w = \sum_{\substack{\mathbf{J}=(J_1,J_2,\ldots,J_k) \text{ is}\\ \text{a set composition of } [n];\\ |J_p|=\gamma_p \text{ for each } p}} \underbrace{\sum_{\substack{w\in S_n;\\ \mathrm{Des}\, w\subseteq \mathrm{Set}(\beta);\\ w(I_p)=J_p \text{ for each } p}} w}_{\substack{=\rho_{\mathbf{J}}\mathbf{B}_{\gamma\to\beta},\\ \text{where } \rho_{\mathbf{J}} \text{ is the unique permutation of } [n]\\ \text{that sends each } I_p \text{ to } J_p \text{ order-preservingly}}}$$

$$= \underbrace{\sum_{\substack{\mathbf{J}=(J_1,J_2,\ldots,J_k) \text{ is}\\ \text{a set composition of } [n];\\ |J_p|=\gamma_p \text{ for each } p}} \rho_{\mathbf{J}}}_{\substack{=\mathbf{B}_\gamma\\ \text{(since the addends } \rho_{\mathbf{J}} \text{ of this sum are}\\ \text{precisely the } w\in S_n \text{ satisfying } \mathrm{Des}\, w\subseteq \mathrm{Set}(\gamma))}} \qquad \mathbf{B}_{\gamma\to\beta} = \mathbf{B}_\gamma \mathbf{B}_{\gamma\to\beta}.$$

Hence $\mathbf{B}_\beta \mathcal{A} = \mathbf{B}_\gamma \mathbf{B}_{\gamma\to\beta} \mathcal{A} \subseteq \mathbf{B}_\gamma \mathcal{A}$. In other words, $\mathcal{R}_\beta \subseteq \mathcal{R}_\gamma$. □

Proposition 1. *If $\beta, \gamma \in \mathrm{Comp}_n$ and $\widetilde{\beta} \preceq_\pi \widetilde{\gamma}$, then $\mathcal{R}_\beta \subseteq \mathcal{R}_\gamma$.*

Proof. Lemma 1 shows that $\mathcal{R}_\beta$ does not change if we permute the entries of β. But this allows us to transform $\widetilde{\beta} \preceq_\pi \widetilde{\gamma}$ into $\beta \preceq \gamma$, and then the claim follows from Lemma 2. □

Hence $(\mathcal{R}_\beta)_{\beta\in\mathrm{Comp}_n}$ is a filtration of $\mathcal{A}$ by right $\mathcal{A}$-submodules that can also be relabelled (removing redundant copies) as $(\mathcal{R}_\beta)_{\beta\in\mathrm{Par}_n}$, where Par_n is the set of all partitions of n. Furthermore, these right ideals are stable under the left action of Σ_n:

Proposition 2. *For every $\beta \in \mathrm{Comp}_n$, the subspace $\mathcal{R}_\beta$ is a $(\Sigma_n, \mathcal{A})$-subbimodule of $\mathcal{A}$.*

Proof. Right $\mathcal{A}$-stability is immediate from the definition. Left-stability follows from Solomon's formula (2.2): for any $\alpha \in \mathrm{Comp}_n$ the product $\mathbf{B}_\alpha \mathbf{B}_\beta$ is a linear combination of $\mathbf{B}_\gamma$ with γ refining β; by Lemma 2 each such $\mathbf{B}_\gamma$ generates a right ideal contained in $\mathcal{R}_\beta$. Therefore $\mathbf{B}_\alpha \mathcal{R}_\beta \subseteq \mathcal{R}_\beta$ for all α, whence $\Sigma_n \cdot \mathcal{R}_\beta \subseteq \mathcal{R}_\beta$. □

We look at the quotient spaces

$$\mathcal{R}_\beta / \sum_{\alpha \prec \beta} \mathcal{R}_\alpha,$$

where the sum runs over compositions α strictly finer than β, or, equivalently, over compositions α satisfying $\widetilde{\alpha} \prec_\pi \widetilde{\beta}$ (the latter sum has more addends than the former, but Lemma 1 shows that these extra addends merely duplicate existing addends). Left multiplication by $\mathbf{B}_\alpha$ acts on these quotient spaces by scalar multiplication:

Theorem 1. *Fix $\alpha, \beta \in \mathrm{Comp}_n$. Let $\eta_\beta(\alpha)$ be the number of functions $f : [\ell(\beta)] \to [\ell(\alpha)]$ such that for every $j \in [\ell(\alpha)]$ one has $\alpha_j = \sum_{i \in f^{-1}(j)} \beta_i$. Then left multiplication by $\mathbf{B}_\alpha$ acts on the quotient $\mathcal{R}_\beta / \sum_{\alpha' \prec \beta} \mathcal{R}_{\alpha'}$ as multiplication by the scalar $\eta_\beta(\alpha)$.*

Proof. This is an easy corollary of Solomon's formula (2.2). Indeed, all matrices M on the right hand side of (2.2) satisfy $\mathrm{read}(M) \preceq \beta$. Among these matrices M, the ones that satisfy $\mathrm{read}(M) = \beta$ are precisely the ones that have exactly one nonzero entry on each row. The number of these latter matrices is $\eta_\beta(\alpha)$, since they can be encoded by the function $f : [\ell(\beta)] \to [\ell(\alpha)]$ where $f(i)$ is the position of the unique nonzero entry in the i-th row of M. Hence, (2.2) becomes

$$\mathbf{B}_\alpha \mathbf{B}_\beta = \eta_\beta(\alpha) \mathbf{B}_\beta + (\text{a sum of } \mathbf{B}_{\alpha'} \text{ with } \alpha' \prec \beta)\,. \tag{2.3}$$

Since $\mathcal{R}_\beta = \mathbf{B}_\beta \mathcal{A}$, this proves the claim. □

Remark 1. *It can be shown that if $\mathbf{k}$ is a $\mathbb{Q}$-algebra, then Proposition 1 can be strengthened as follows: If $\beta, \gamma \in \mathrm{Comp}_n$ and $\widetilde{\beta} \preceq_\pi \widetilde{\gamma}$, then $\mathbf{B}_\beta \Sigma_n \subseteq \mathbf{B}_\gamma \Sigma_n$. However, our Proposition 1 holds independently of $\mathbf{k}$.*

3 The left-to-right minima basis of $\mathcal{A}$

3.1 Left-to-right minima of a permutation

We begin by recalling a classical statistic on permutations.

Definition 4 (Left-to-right minima)**.** *Let $w \in S_n$. The* left-to-right minima *(short:* LRMs*) of w are the entries*

$$\mathrm{LRM}(w) := \Big\{ i \in [n] \;\Big|\; \forall\, k < w^{-1}(i),\;\; w(k) > i \Big\}.$$

Thus i belongs to $\mathrm{LRM}(w)$ if it is smaller than every entry appearing to its left in the one-line notation of w. Clearly, $\mathrm{LRM}(w)$ always contains 1 and $w(1)$. We also define the shifted set

$$\mathrm{LRM}'(w) := \{\, \ell - 1 \mid \ell \in \mathrm{LRM}(w),\; \ell > 1 \} \subseteq [n-1],$$

and finally the associated composition (see §2.1 for notations)

$$\mathrm{cLRM}'(w) := \mathrm{Comp}\Big(\mathrm{LRM}'(w)\Big).$$

The following easy fact relates the LRMs of a permutation to those of its inverse.

Remark 2. *For every $w \in S_n$, one has $\mathrm{LRM}(w) = w\Big(\mathrm{LRM}(w^{-1})\Big)$.*

Example 1. *Lemma 2 says that the LRMs of w^{-1} occur at the* LRM positions *of w. For example, if $w = \mathbf{6}7\mathbf{2}49\mathbf{1}853$, then $\mathrm{LRM}(w) = \{6, 2, 1\}$, whereas $w^{-1} = \mathbf{6}\mathbf{3}948\mathbf{1}275$ and $\mathrm{LRM}(w^{-1}) = \{6, 3, 1\}$, which matches the fact that the LRMs of w appear at positions $6, 3, 1$.*

3.2 A new basis of the group algebra

For each permutation $w \in S_n$ we consider the element

$$\mathbf{B}_{\mathrm{LRM}'(w)}\, w \ \in \ \mathcal{A}.$$

We shall prove that these vectors form a basis of $\mathcal{A}$. Two lemmas will guide our way.

Lemma 3. *Let $u \in S_n$ be a permutation that has at least one descent. Let m be the largest descent of u. Let $\ell = m+1$. Then:*

(a) *We have $u(\ell) < \ell$.*

(b) *We have $u(i) \leq i$ for all $i > \ell$.*

Proof. **(a)** Since $\ell - 1 = m$ is the largest descent of u, we have

$$u(\ell-1) > u(\ell) < u(\ell+1) < u(\ell+2) < \cdots < u(n). \tag{3.1}$$

Thus, the $n-\ell+1$ distinct numbers $u(\ell-1),\ u(\ell+1),\ u(\ell+2),\ \ldots,\ u(n)$ are all larger than $u(\ell)$, hence belong to the interval $[u(\ell)+1,\ n]$. Therefore, this interval must contain at least $n-\ell+1$ numbers. In other words, $|[u(\ell)+1,\ n]| \geq n-\ell+1 > n-\ell$. Since $|[u(\ell)+1,\ n]| = n - u(\ell)$, this rewrites as $n - u(\ell) > n - \ell$, whence $u(\ell) < \ell$. This proves part **(a)**.

(b) Let $i > \ell$. Then, (3.1) yields $u(i) < u(i+1) < \cdots < u(n)$. Thus, the $n-i$ distinct numbers $u(i+1),\ u(i+2),\ \ldots,\ u(n)$ are all larger than $u(i)$, hence belong to the interval $[u(i)+1,\ n]$. Therefore, this interval must contain at least $n-i$ numbers. In other words, $|[u(i)+1,\ n]| \geq n-i$. Since $|[u(i)+1,\ n]| = n - u(i)$, this rewrites as $n - u(i) \geq n - i$, whence $u(i) < i$. This proves part **(b)**. □

Lemma 4. *Let $\leq_{\mathrm{lex}}$ denote the lexicographic order relation on permutations seen as their one-line notation. Let $w, u \in S_n$ be such that $\mathrm{Des}(u) \subseteq \mathrm{LRM}'(w)$. Then, $uw \ \leq_{\mathrm{lex}}\ w$.*

Proof. Assume that u has at least one descent (since otherwise, we have $u(1) \leq u(2) \leq \cdots \leq u(n)$, therefore $u = \mathrm{id}$ and thus $uw = w$, rendering the lemma obvious). Let m be the largest descent of u. Thus, $m \in \mathrm{Des}(u)$. Let $\ell = m+1$. Then, Lemma 3 **(a)** yields $u(\ell) < \ell$. Furthermore, $\ell - 1 = m \in \mathrm{Des}(u) \subseteq \mathrm{LRM}'(w)$, hence $\ell \in \mathrm{LRM}(w)$. Therefore, every entry i appearing to the left of ℓ in the one-line notation of w is $> \ell$, and thus satisfies $u(i) \leq i$ by Lemma 3 **(b)**. Combining this with $u(\ell) < \ell$, we see that $uw <_{\mathrm{lex}} w$ lexicographically. This proves Lemma 4. □

Corollary 1. *For every $w \in S_n$, we have*

$$\mathbf{B}_{\mathrm{LRM}'(w)}\, w = w + \Big(\textit{a } \mathbf{k}\textit{-linear combination of lexicographically smaller permutations}\Big).$$

Proof. Expand:

$$\mathbf{B}_{\mathrm{LRM}'(w)} = \sum_{\substack{u \in S_n; \\ \mathrm{Des}(u) \subseteq \mathrm{LRM}'(w)}} u, \qquad \mathbf{B}_{\mathrm{LRM}'(w)}\, w = \sum_{\substack{u \in S_n; \\ \mathrm{Des}(u) \subseteq \mathrm{LRM}'(w)}} uw.$$

By Lemma 4, every summand uw is lexicographically smaller than w, and equality occurs only for $u = \mathrm{id}$. Thus the claim follows. □

Corollary 2. *The family*

$$\left\{ \mathbf{B}_{\mathrm{LRM}'(w)}\, w \;\middle|\; w \in S_n \right\}$$

is a basis of $\mathcal{A}$.

Proof. Corollary 1 shows that this family expands unitriangularly in the standard basis $\{w\}_{w \in S_n}$. □

Example 2. *Let us write permutations* $\sigma \in S_n$ *in one-line notation, i.e., as* $[\sigma(1)\ \sigma(2)\ \cdots\ \sigma(n)]$. *For* $n = 3$, *the basis* $\left\{ \mathbf{B}_{\mathrm{LRM}'(w)}\, w \;\middle|\; w \in S_n \right\}$ *of* $\mathcal{A}$ *consists of the vectors*

$$\begin{aligned}
&\mathbf{B}_{\mathrm{LRM}'([123])}\ [123] = \mathbf{B}_{(3)}\ [123] = [123]\,; \\
&\mathbf{B}_{\mathrm{LRM}'([132])}\ [132] = \mathbf{B}_{(3)}\ [132] = [132]\,; \\
&\mathbf{B}_{\mathrm{LRM}'([213])}\ [213] = \mathbf{B}_{(1,2)}\ [213] = [213] + [123] + [132]\,; \\
&\mathbf{B}_{\mathrm{LRM}'([231])}\ [231] = \mathbf{B}_{(1,2)}\ [231] = [231] + [132] + [123]\,; \\
&\mathbf{B}_{\mathrm{LRM}'([312])}\ [312] = \mathbf{B}_{(2,1)}\ [312] = [312] + [213] + [123]\,; \\
&\mathbf{B}_{\mathrm{LRM}'([321])}\ [321] = \mathbf{B}_{(1,1,1)}\ [321] = [321] + [312] + [231] + [213] + [132] + [123]\,.
\end{aligned}$$

3.3 Connection with the B-basis filtration

We now relate this new basis to the filtration introduced in Section 2. Recall the filtration subspaces $\mathcal{R}_\alpha$ indexed by compositions.

Theorem 2. *Let* $\alpha \in \mathrm{Comp}_n$ *and define*

$$\mathcal{S}_\alpha := \mathrm{span}\left\{ \mathbf{B}_{\mathrm{LRM}'(w)}\, w \;\middle|\; \widetilde{\mathrm{cLRM}'(w)} \preceq_\pi \widetilde{\alpha} \right\}.$$

Then

$$\mathcal{S}_\alpha = \mathcal{R}_\alpha.$$

The inclusion $\mathcal{S}_\alpha \subseteq \mathcal{R}_\alpha$ in Theorem 2 follows directly from Proposition 1. The converse inclusion $\mathcal{R}_\alpha \subseteq \mathcal{S}_\alpha$ is much harder, and will be proved in §4.

Remark 3. *The use of $\preceq_\pi$ (rather than $\preceq$) is essential. The seemingly more natural span $\mathcal{S}'_\alpha := \operatorname{span}\left\{\mathbf{B}_{\mathrm{LRM}'(w)}\, w \;\middle|\; \mathrm{cLRM}'(w) \preceq \alpha\right\}$ is often smaller than $\mathcal{R}_\alpha$. For instance,*

for $n = 3$:

α	$(1,1,1)$	$(1,2)$	$(2,1)$	(3)
$\dim\left(\mathcal{R}_\alpha\right)$	1	4	4	6
$\dim\left(\mathcal{S}'_\alpha\right)$	1	3	2	6

Combining Theorem 1 with Theorem 2, we obtain:

Corollary 3. *(This is an outline; see Section 6 for a more detailed statement.)*

Let $\mathbf{a} = \sum_{\alpha\in\mathrm{Comp}_n} \lambda_\alpha \mathbf{B}_\alpha$ (with $\lambda_\alpha \in \mathbf{k}$) be an arbitrary element of the descent algebra Σ_n. Then, left multiplication by $\mathbf{a}$ on $\mathcal{A}$ (that is, the linear map $\mathcal{A} \to \mathcal{A}$, $\mathbf{x} \mapsto \mathbf{a}\mathbf{x}$) is represented with respect to the basis from Corollary 2 by a triangular matrix (triangular with respect to an appropriate order on Comp_n). The diagonal entries of this matrix (and thus the eigenvalues of this map, listed with their algebraic multiplicities) are the numbers $\sum_{\alpha\in\mathrm{Comp}_n} \lambda_\alpha \eta_{\mathrm{cLRM}'(w)}(\alpha)$, where w ranges over S_n, and where we use the $\eta_\beta(\alpha)$ defined in Theorem 1.

The mere **existence** of such a basis when $\mathbf{k}$ is a field is not new, and is implicit in the works of Bidigare, Brown and others. The formula for the eigenvalues can be pieced together from [5, Proposition 4.2] and [1, Corollary 4.1.3] and Foata's fundamental transformation. Our innovation is constructing such a basis explicitly, for all commutative rings $\mathbf{k}$; this yields a wholly new proof of Corollary 3.

4 The free algebra and the Dynkin elements

4.1 Definitions

We now introduce the main ingredients in the proof of Theorem 2.

Let W_n be the monoid of all *words* (i.e., finite tuples) with letters (i.e., entries) in $[n]$; its multiplication is concatenation. We denote its generators – i.e., the letters – by $\underline{1}, \underline{2}, \ldots, \underline{n}$, so as to distinguish them from the respective numbers. The monoid algebra $F_n := \mathbf{k}[W_n]$ of W_n is the ring of *noncommutative polynomials* (aka the *free algebra*) in n indeterminates $\underline{1}, \underline{2}, \ldots, \underline{n}$ over $\mathbf{k}$. It is graded by degree (i.e., word length), and we let $F_{n,k}$ denote its k-th graded component, with basis given by the set $W_{n,k}$ of length-k words. The symmetric group S_k acts on $W_{n,k}$ from the right by permuting the letters:

$$\left(\underline{w_1}\ \underline{w_2}\ \cdots\ \underline{w_k}\right)\cdot\sigma = \underline{w_{\sigma(1)}}\ \underline{w_{\sigma(2)}}\ \cdots\ \underline{w_{\sigma(k)}} \qquad \text{for } w_i \in [n] \text{ and } \sigma \in S_k.$$

Extending this action linearly, we obtain a right $\mathbf{k}[S_k]$-action on $\mathbf{k}[W_{n,k}] = F_{n,k}$. Two words in $W_{n,k}$ are said to be *anagrams* (of each other) if they belong to the same S_k-orbit.

There is an injective map $S_n \to W_{n,n}$, sending each permutation $\sigma \in S_n$ to its one-line notation $\underline{\sigma} := \underline{\sigma(1)}\ \underline{\sigma(2)}\ \cdots\ \underline{\sigma(n)}$. By linearity, it extends to an injective $\mathbf{k}$-linear map $\mathbf{k}[S_n] \to \mathbf{k}[W_{n,n}] = F_{n,n}$, which we also write as $a \mapsto \underline{a}$. This map is not a ring morphism, but it respects the right $\mathbf{k}[S_n]$-actions, i.e., satisfies

$$\underline{a} \cdot b = \underline{ab} \qquad \text{for each } a, b \in \mathbf{k}[S_n]. \tag{4.1}$$

A word $w = (w_1, w_2, \ldots, w_k)$ is said to have *V-shape* if it satisfies

$$w_1 > w_2 > \cdots > w_i < w_{i+1} < w_{i+2} < \cdots < w_k \qquad \text{for some } i \in [k]$$

(that is, it first decreases, then increases). In this case, the i is unique and is denoted $\operatorname{val} w$. A permutation $v \in S_n$ is said to have *V-shape* on an interval $[i,j] = \{i, i+1, \ldots, j\}$ if the word $(v(i), v(i+1), \ldots, v(j))$ has V-shape.

When $a, b \in F_n$, we write $[a,b]$ for the *commutator* $ab - ba$.

The crucial notion to us is the following:

Definition 5. *For any nonempty subset S of $[n]$, we define the* Dynkin element *(or* nested commutator*)*

$$V^S := \Big[\Big[\cdots\Big[\Big[\underline{s_1},\ \underline{s_2}\Big],\ \underline{s_3}\Big],\ \ldots\Big],\ \underline{s_k}\Big],$$

where $s_1, s_2, \ldots, s_k$ are the elements of S in increasing order. (If $k = 1$, then this means just $\underline{s_1}$.)

Example 3. *We have*

$$V^{\{2,6,7\}} = [[\underline{2},\ \underline{6}],\ \underline{7}] = \underline{2}\ \underline{6}\ \underline{7} - \underline{6}\ \underline{2}\ \underline{7} - \underline{7}\ \underline{2}\ \underline{6} + \underline{7}\ \underline{6}\ \underline{2};$$
$$V^{\{1,4\}} = [\underline{1},\ \underline{4}] = \underline{1}\ \underline{4} - \underline{4}\ \underline{1}; \qquad V^{\{3\}} = \underline{3}.$$

The letter V is chosen because of the following simple formula:

Proposition 3. *Let S be a nonempty subset of $[n]$. Then,*

$$V^S = \sum (-1)^{\operatorname{val} w - 1}\, w,$$

where the sum ranges over all words w that contain each element of S exactly once (and no other letters) and have V-shape.

Proof. Easy induction on $|S|$, since $V^S = \left[V^{S \setminus \{\max S\}},\ \underline{\max S}\right]$ for $|S| > 1$. □

Dynkin elements are fundamental to the study of the free Lie algebra; see, e.g., [6, (8.4.2)]. We shall have more use for their products in F_n.

Definition 6. *Let $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_p) \in \operatorname{Comp}(n)$ be a composition of n. Then, we set*

$$\mathbf{V}_\alpha = V^{\operatorname{Set}(\alpha)_1} V^{\operatorname{Set}(\alpha)_2} \cdots V^{\operatorname{Set}(\alpha)_p}.$$

Example 4. *For the compositions $(2,1)$ and $(2,2)$ we obtain:*

$$\mathbf{V}_{(2,1)} = V^{\{1,2\}} V^{\{3\}} = (\underline{1}\ \underline{2} - \underline{2}\ \underline{1})\ \underline{3} = \underline{1}\ \underline{2}\ \underline{3} - \underline{2}\ \underline{1}\ \underline{3};$$
$$\mathbf{V}_{(2,2)} = V^{\{1,2\}} V^{\{3,4\}} = (\underline{1}\ \underline{2} - \underline{2}\ \underline{1})(\underline{3}\ \underline{4} - \underline{4}\ \underline{3}) = \underline{1}\ \underline{2}\ \underline{3}\ \underline{4} - \underline{1}\ \underline{2}\ \underline{4}\ \underline{3} - \underline{2}\ \underline{1}\ \underline{3}\ \underline{4} + \underline{2}\ \underline{1}\ \underline{4}\ \underline{3}.$$

4.2 Action on the B-basis

The Dynkin elements $\mathbf{V}_\alpha$ are themselves images of certain elements of the descent algebra Σ_n under the injection $\mathbf{k}[S_n] \to \mathbf{k}[W_{n,n}] = F_{n,n},\ a \mapsto \underline{a}$, namely of the "*power sums of the first kind*" Ψ_α studied in [4, §5]. However, we will use another of their properties – namely, a formula for the action of $\mathbf{B}_\beta \in \mathbf{k}[S_n]$ on the $\mathbf{V}_\alpha$, which appears as Lemma 9.33 in [6] and as Theorem 2.1 in [3]:

Theorem 3 (Lemma 9.33 of [6], or Theorem 2.1 of [3])**.** *Let $\beta = (\beta_1, \beta_2, \ldots, \beta_p)$ and $\gamma = (\gamma_1, \gamma_2, \ldots, \gamma_q)$ be two compositions of n. Let $\mathcal{L}_{\beta,\gamma}$ be the set of all ordered set partitions $T = (T_1, T_2, \ldots, T_q)$ of the index set $[p] = \{1, 2, \ldots, p\}$ into q blocks that satisfy*

$$\sum_{i \in T_j} \beta_i = \gamma_j \quad \text{for all } j. \tag{4.2}$$

For each block T_j of T, let $V^{!T_j}$ denote the product of the Dynkin elements $V^{\mathrm{Set}(\beta)_i}$ with $i \in T_j$ (in the order of increasing i). Then

$$\mathbf{V}_\beta \mathbf{B}_\gamma = \sum_{T=(T_1,T_2,\ldots,T_q) \in \mathcal{L}_{\beta,\gamma}} V^{!T_1} V^{!T_2} \cdots V^{!T_q}. \tag{4.3}$$

Remark 4. *Reutenauer's [6, Lemma 9.33] is even more general: He views F_n as the tensor algebra $T(V)$ of the free $\mathbf{k}$-module $V = \mathbf{k}^n$ with basis $(\underline{1}, \underline{2}, \ldots, \underline{n})$. This equips F_n with a Hopf algebra structure: that of the tensor Hopf algebra $T(V)$. All Dynkin elements V^S are primitive elements of $T(V)$, as can be easily proved by induction on $|S|$.*

Let $\gamma = (\gamma_1, \gamma_2, \ldots, \gamma_q)$ be a composition of n, and let $P_1, P_2, \ldots, P_k$ be k homogeneous primitive elements of $T(V)$ with total degree $\deg P_1 + \deg P_2 + \cdots + \deg P_k = n$.

Set $P_W := P_{w_1} P_{w_2} \cdots P_{w_m}$ for any subset $W = \{w_1 < w_2 < \cdots < w_m\}$ of $[k]$.

Then, the right action of the symmetric group S_n on $F_{n,n}$ satisfies

$$(P_1 P_2 \cdots P_k)\, \mathbf{B}_\gamma = \sum P_{T_1} P_{T_2} \cdots P_{T_q},$$

where the sum ranges over all ordered set partitions $(T_1, T_2, \ldots, T_q)$ of $[k]$ such that $\deg P_{T_i} = \gamma_i$ for each i.

This can be proved slickly using Hopf algebra theory; here is an outline of the proof:

1. *First, observe that the action of $\mathbf{B}_\gamma$ on the n-th degree component of $T(V)$ is the Hopf-algebraic operator $m^{[q]} \circ P_\gamma \circ \Delta^{[q]}$ (where $\Delta^{[q]} : T(V) \to T(V)^{\otimes q}$ is the q-fold comultiplication, $m^{[q]} : T(V)^{\otimes q} \to T(V)$ is the q-fold multiplication, and P_γ is the projection $T(V)^{\otimes q} \to T(V)^{\otimes q}$ onto the $(\gamma_1, \gamma_2, \ldots, \gamma_q)$-th multigraded component), since $\Delta^{[q]}$ sends each "word" $v_1 v_2 \cdots v_n \in T(V)$ (with $v_1, v_2, \ldots, v_n \in V$) to the tensor $\sum_{W_1 \sqcup W_2 \sqcup \cdots \sqcup W_q = [n]} v_{W_1} \otimes v_{W_2} \otimes \cdots \otimes v_{W_q} \in T(V)^{\otimes q}$, where v_W for a subset $W = \{w_1 < w_2 < \cdots < w_m\} \subseteq [n]$ is defined to be $v_{w_1} v_{w_2} \cdots v_{w_m} \in T(V)$.*

2. *Then, show that the latter operator $m^{[q]} \circ P_\gamma \circ \Delta^{[q]}$ sends $P_1 P_2 \cdots P_k$ to $\sum P_{T_1} P_{T_2} \cdots P_{T_q}$. The easiest way to do so is by observing that this holds when $P_1, P_2, \ldots, P_k$ are themselves among the generators of the free algebra $T(V)$ (that is, belong to a basis of V), but the general case then follows easily (since the primitivity of $P_1, P_2, \ldots, P_k$ ensures that there is a bialgebra morphism from a free algebra to $T(V)$ that sends its generators to $P_1, P_2, \ldots, P_k$).*

Remark 5. *The products $V^{!T_1} V^{!T_2} \cdots V^{!T_q}$ in Theorem 3 are – in general – not of the form $\mathbf{V}_\gamma$. For instance,*

$$V^{\{3\}} V^{\{1,2\}} = \underline{3}(\underline{1}\ \underline{2} - \underline{2}\ \underline{1}) = \underline{3}\ \underline{1}\ \underline{2} - \underline{3}\ \underline{2}\ \underline{1},$$

a summand in $\mathbf{V}_{(2,1)} \mathbf{B}_{(1,2)}$, differs from both

$$\mathbf{V}_{(2,1)} = V^{\{1,2\}} V^{\{3\}} = (\underline{1}\ \underline{2} - \underline{2}\ \underline{1})\underline{3} = \underline{1}\ \underline{2}\ \underline{3} - \underline{2}\ \underline{1}\ \underline{3}$$

and

$$\mathbf{V}_{(1,2)} = \underline{1}(\underline{2}\ \underline{3} - \underline{3}\ \underline{2}) = \underline{1}\ \underline{2}\ \underline{3} - \underline{1}\ \underline{3}\ \underline{2}.$$

Example 5. *Consider $\alpha = (2,2)$ and $\beta = (3,1)$. We have*

$$\begin{aligned} \mathbf{V}_{(2,2)} &= (\underline{12} - \underline{21})(\underline{34} - \underline{43}) = \underline{1234} - \underline{2134} - \underline{1243} + \underline{2143}, \\ \mathbf{B}_{(3,1)} &= 1234 + 1243 + 1342 + 2341. \end{aligned}$$

Hence

$$\begin{aligned} \mathbf{V}_{(2,2)} \mathbf{B}_{(3,1)} &= (\underline{1234} - \underline{2134} - \underline{1243} + \underline{2143})(1234 + 1243 + 1342 + 2341) \\ &= \underline{1234} - \underline{2134} - \underline{1243} + \underline{2143} + \underline{1243} - \underline{2143} - \underline{1234} + \underline{2134} \\ &\qquad + \underline{1342} - \underline{2341} - \underline{1432} + \underline{2431} + \underline{2341} - \underline{1342} - \underline{2431} + \underline{1432} \\ &= 0. \end{aligned}$$

On the other hand, for $\alpha = \beta = (2,2)$,

$$\begin{aligned} \mathbf{V}_{(2,2)} \mathbf{B}_{(2,2)} &= (\underline{1234} - \underline{2134} - \underline{1243} + \underline{2143})(1234 + 1423 + 1324 + 2413 + 2314 + 3412) \\ &= V_2^{\{1,2\}} V_2^{\{3,4\}} + V_2^{\{3,4\}} V_2^{\{1,2\}} \neq 0. \end{aligned}$$

5 Proof of the main theorem

A direct consequence of Theorem 3 is the following vanishing criterion:

$$\widetilde{\beta} \not\leq_\pi \widetilde{\gamma} \quad \Longrightarrow \quad \mathbf{V}_\beta \mathbf{B}_\gamma = 0 \tag{5.1}$$

(since $\mathcal{L}_{\beta,\gamma} = \varnothing$ in this case). A slightly less obvious particular case is the following:

Lemma 5. *Let $\beta = (\beta_1, \beta_2, \ldots, \beta_p)$ and $\gamma = (\gamma_1, \gamma_2, \ldots, \gamma_p)$ be two compositions of n having the same length. Then,*

$$\mathbf{V}_\beta \mathbf{B}_\gamma = \sum_{\substack{\chi \in S_p; \\ \beta_{\chi(i)} = \gamma_i \text{ for each } i}} V^{\mathrm{Set}(\beta)_{\chi(1)}} V^{\mathrm{Set}(\beta)_{\chi(2)}} \cdots V^{\mathrm{Set}(\beta)_{\chi(p)}}.$$

Proof. Set $q = p$, and apply Theorem 3. Observe that any ordered set partition $T \in \mathcal{L}_{\beta,\gamma}$ must consist entirely of 1-element blocks (since it partitions a p-element set into p blocks), and thus has the form $(\{\chi(1)\}, \{\chi(2)\}, \ldots, \{\chi(p)\})$ for a unique permutation $\chi \in S_p$. Moreover, this latter permutation χ must satisfy $\beta_{\chi(i)} = \gamma_i$ for all $i \in [p]$, in order for T to satisfy (4.2). This is necessary and sufficient for $T \in \mathcal{L}_{\beta,\gamma}$. Thus, (4.3) rewrites as

$$\mathbf{V}_\beta \mathbf{B}_\gamma = \sum_{\substack{\chi \in S_p; \\ \beta_{\chi(i)} = \gamma_i \text{ for each } i}} V^{!\{\chi(1)\}} \cdots V^{!\{\chi(p)\}} = \sum_{\substack{\chi \in S_p; \\ \beta_{\chi(i)} = \gamma_i \text{ for each } i}} V^{\mathrm{Set}(\beta)_{\chi(1)}} \cdots V^{\mathrm{Set}(\beta)_{\chi(p)}}.$$

This proves Lemma 5. □

We shall now study the products $V^{\mathrm{Set}(\beta)_{\chi(1)}} V^{\mathrm{Set}(\beta)_{\chi(2)}} \cdots V^{\mathrm{Set}(\beta)_{\chi(p)}}$ on the right hand side of Lemma 5 more closely. These products are multilinear in the generators $\underline{1}, \underline{2}, \ldots, \underline{n}$ of F_n, so they are linear combinations of words $\underline{v}$ with $v \in S_n$. The next lemma reveals some insights about the v's that appear in these combinations.

Lemma 6. *Let $w, u \in S_n$. Assume that the compositions $\beta = (\beta_1, \beta_2, \ldots, \beta_p) := \mathrm{cLRM}'(w)$ and $\gamma = (\gamma_1, \gamma_2, \ldots, \gamma_p) := \mathrm{cLRM}'(u)$ are anagrams of one another, and let $\chi \in S_p$ be a permutation satisfying*

$$\beta_{\chi(i)} = \gamma_i \qquad \text{for each } i \in [p]. \tag{5.2}$$

Let $v \in S_n$ be a permutation such that the word $\underline{v} \in W_{n,n}$ appears in the expansion of the product $V^{\mathrm{Set}(\beta)_{\chi(1)}} \cdots V^{\mathrm{Set}(\beta)_{\chi(p)}}$. Equivalently, let $v \in S_n$ be a permutation that sends each $\mathrm{Set}(\gamma)_i$ to $\mathrm{Set}(\beta)_{\chi(i)}$ and has V-shape on each $\mathrm{Set}(\gamma)_i$ (this is equivalent by Proposition 3 and by (5.2)). Assume that

$$vu = w.$$

Then, $\chi = \mathrm{id}$ and $v = \mathrm{id}$ and $\gamma = \beta$.

Proof. For each $i \in [p]$, we define a claim $\mathcal{C}(i)$ as follows:

> *Claim $\mathcal{C}(i)$:* We have $\chi(i) = i$ and $\gamma_i = \beta_i$ and $\mathrm{Set}(\beta)_i = \mathrm{Set}(\gamma)_i$ and $v\mid_{\mathrm{Set}(\beta)_i} = \mathrm{id}$.

It clearly suffices to show that $\mathcal{C}(i)$ holds for all $i \in [p]$. We will prove this by strong descending induction on i. That is, we fix $i \in [p]$, and we assume that all the "higher" claims $\mathcal{C}(i+1), \mathcal{C}(i+2), \ldots, \mathcal{C}(p)$ hold. We must then prove $\mathcal{C}(i)$.

By (2.1), we have $\mathrm{Set}(\beta)_i = [\beta_{\leq i-1} + 1,\ \beta_{\leq i}]$ and $\mathrm{Set}(\gamma)_i = [\gamma_{\leq i-1} + 1,\ \gamma_{\leq i}]$.

If $\sigma \in S_n$ is a permutation and $j \leq |\operatorname{LRM}\sigma|$ is a positive integer, then we let $\operatorname{lrm}_j \sigma$ denote the j-th largest LRM of σ. We extend this to $j = 0$ by setting $\operatorname{lrm}_0 \sigma := n+1$. We note that the LRMs of a permutation σ appear in decreasing order when reading σ from left to right, so that $\operatorname{lrm}_j \sigma$ is the j-th LRM of σ encountered in such a read. Thus, it is easy to see that each $\sigma \in S_n$ and each positive $j \leq |\operatorname{LRM}\sigma|$ satisfy

$$\operatorname{lrm}_j \sigma = \sigma\left(\min\left(\sigma^{-1}\left(\left[\operatorname{lrm}_{j-1}\sigma - 1\right]\right)\right)\right). \tag{5.3}$$

(Indeed, this is just saying that if we remove all numbers larger than $\operatorname{lrm}_{j-1}\sigma - 1$ from the one-line notation of σ, then the first of the remaining numbers will be $\operatorname{lrm}_j \sigma$; but this is clear from the definition of an LRM.)

But $\beta = \operatorname{cLRM}'(w)$ shows that $\beta_{\leq i-1} + 1$ is the i-th smallest LRM of w, and hence is the $(p+1-i)$-th largest LRM of w (since w has $\ell\left(\operatorname{cLRM}'(w)\right) = \ell(\beta) = p$ many LRMs in total). That is, $\beta_{\leq i-1} + 1 = \operatorname{lrm}_{p+1-i} w$. Likewise, $\beta_{\leq i} + 1 = \operatorname{lrm}_{p-i} w$. But (5.3) (applied to $\sigma = w$ and $j = p+1-i$) yields

$$\operatorname{lrm}_{p+1-i} w = w\left(\min\left(w^{-1}\left(\left[\operatorname{lrm}_{p-i} w - 1\right]\right)\right)\right).$$

Using $\beta_{\leq i-1} + 1 = \operatorname{lrm}_{p+1-i} w$ and $\beta_{\leq i} + 1 = \operatorname{lrm}_{p-i} w$, this rewrites as

$$\beta_{\leq i-1} + 1 = w\left(\min\left(w^{-1}\left(\left[\beta_{\leq i}\right]\right)\right)\right). \tag{5.4}$$

Similarly,

$$\gamma_{\leq i-1} + 1 = u\left(\min\left(u^{-1}\left(\left[\gamma_{\leq i}\right]\right)\right)\right). \tag{5.5}$$

However, our induction hypothesis yields $\beta_j = \gamma_j$ for all $j > i$, and thus easily $\beta_{\leq i} = \gamma_{\leq i}$ (since $|\beta| = n = |\gamma|$). Furthermore, it shows that $v\left|_{\operatorname{Set}(\beta)_j}\right. = \operatorname{id}$ for all $j > i$, and thus $v\left|_{[n]-[\beta_{\leq i}]}\right. = \operatorname{id}$, so that $v\left([n] - [\beta_{\leq i}]\right) = [n] - [\beta_{\leq i}]$. By taking complements, we obtain $v\left([\beta_{\leq i}]\right) = [\beta_{\leq i}]$, thus $v^{-1}\left([\beta_{\leq i}]\right) = [\beta_{\leq i}]$. But $vu = w$, so that $w^{-1} = u^{-1}v^{-1}$ and therefore $w^{-1}\left([\beta_{\leq i}]\right) = u^{-1}\left(v^{-1}\left([\beta_{\leq i}]\right)\right) = u^{-1}\left([\beta_{\leq i}]\right)$, because $v^{-1}\left([\beta_{\leq i}]\right) = [\beta_{\leq i}]$.

Now, applying v to (5.5), we find

$$\begin{aligned}
v\left(\gamma_{\leq i-1} + 1\right) &= v\left(u\left(\min\left(u^{-1}\left(\left[\gamma_{\leq i}\right]\right)\right)\right)\right) \\
&= w\left(\min\left(u^{-1}\left(\left[\beta_{\leq i}\right]\right)\right)\right) && \left(\text{since } vu = w \text{ and } \gamma_{\leq i} = \beta_{\leq i}\right) \\
&= w\left(\min\left(w^{-1}\left(\left[\beta_{\leq i}\right]\right)\right)\right) && \left(\text{since } w^{-1}\left(\left[\beta_{\leq i}\right]\right) = u^{-1}\left(\left[\beta_{\leq i}\right]\right)\right) \\
&= \beta_{\leq i-1} + 1 \qquad (\text{by (5.4)}) && (5.6) \\
&\in \left[\beta_{\leq i-1} + 1,\ \beta_{\leq i}\right] = \operatorname{Set}(\beta)_i. && (5.7)
\end{aligned}$$

But $\gamma_{\leq i-1} + 1 \in \left[\gamma_{\leq i-1} + 1,\ \gamma_{\leq i}\right] = \operatorname{Set}(\gamma)_i$, so that

$$v\left(\gamma_{\leq i-1} + 1\right) \in v\left(\operatorname{Set}(\gamma)_i\right) = \operatorname{Set}(\beta)_{\chi(i)}$$

(by the requirements on v). Confronting this with (5.7), we conclude that $\chi(i) = i$, since otherwise $\operatorname{Set}(\beta)_{\chi(i)}$ would be disjoint from $\operatorname{Set}(\beta)_i$. Thus, (5.2) becomes $\gamma_i = \beta_i$. Thus, $\gamma_{\leq i-1} = \beta_{\leq i-1}$ (by subtracting $\gamma_i = \beta_i$ from $\gamma_{\leq i} = \beta_{\leq i}$). Combining this with $\gamma_{\leq i} = \beta_{\leq i}$, we conclude that $\operatorname{Set}(\beta)_i = \operatorname{Set}(\gamma)_i$.

Recall that the permutation v has V-shape on $\operatorname{Set}(\gamma)_i = [\gamma_{\leq i-1}+1,\ \gamma_{\leq i}]$. That is, the word $(v(\gamma_{\leq i-1}+1), v(\gamma_{\leq i-1}+2), \ldots, v(\gamma_{\leq i}))$ has V-shape. However, the letters of this word are $\gamma_{\leq i-1}+1, \gamma_{\leq i-1}+2, \ldots, \gamma_{\leq i}$ in some order (since v sends $\operatorname{Set}(\gamma)_i$ to $\operatorname{Set}(\beta)_{\chi(i)} = \operatorname{Set}(\beta)_i = \operatorname{Set}(\gamma)_i$), and moreover the **leftmost** of these letters is the **smallest** of them (since $\gamma_{\leq i-1} = \beta_{\leq i-1}$ allows us to rewrite (5.6) as $v(\gamma_{\leq i-1}+1) = \gamma_{\leq i-1}+1$). But a word that has V-shape and starts with its smallest letter must be increasing; thus, in particular, the word $(v(\gamma_{\leq i-1}+1), v(\gamma_{\leq i-1}+2), \ldots, v(\gamma_{\leq i}))$ is increasing. Since its letters are $\gamma_{\leq i-1}+1, \gamma_{\leq i-1}+2, \ldots, \gamma_{\leq i}$, we thus obtain

$$(v(\gamma_{\leq i-1}+1), v(\gamma_{\leq i-1}+2), \ldots, v(\gamma_{\leq i})) = (\gamma_{\leq i-1}+1, \gamma_{\leq i-1}+2, \ldots, \gamma_{\leq i}).$$

In other words, $v|_{\operatorname{Set}(\gamma)_i} = \mathrm{id}$. That is, $v|_{\operatorname{Set}(\beta)_i} = \mathrm{id}$ (since $\operatorname{Set}(\beta)_i = \operatorname{Set}(\gamma)_i$). We have now proved all four parts of our claim $\mathcal{C}(i)$. Thus, the induction step is complete, and as we said, this finishes the proof of the lemma. □

For any $a \in F_n$ and any $\mathfrak{w} \in W_n$, let $[\mathfrak{w}](a)$ denote the $\mathfrak{w}$-coefficient of a.

Lemma 7. *Let $w, u \in S_n$. Set $\beta := \mathrm{cLRM}'(w)$ and $\gamma := \mathrm{cLRM}'(u)$. Then:*

(a) *We have $[\underline{w}](\mathbf{V}_\beta \mathbf{B}_\beta w) = 1$.*

(b) *If $u \neq w$, then $[\underline{w}](\mathbf{V}_\beta \mathbf{B}_\gamma u) = 0$ unless $\widetilde{\beta} \prec_\pi \widetilde{\gamma}$.*

Proof. **(a)** Write β as $\beta = (\beta_1, \beta_2, \ldots, \beta_p)$. Applying Lemma 5 to $\gamma = \beta$, we find

$$\mathbf{V}_\beta \mathbf{B}_\beta = \sum_{\substack{\chi \in S_p; \\ \beta_{\chi(i)} = \beta_i \text{ for each } i}} V^{\operatorname{Set}(\beta)_{\chi(1)}} \cdots V^{\operatorname{Set}(\beta)_{\chi(p)}}.$$

Hence,

$$[\underline{w}](\mathbf{V}_\beta \mathbf{B}_\beta w) = \sum_{\substack{\chi \in S_p; \\ \beta_{\chi(i)} = \beta_i \text{ for each } i}} [\underline{w}]\left(V^{\operatorname{Set}(\beta)_{\chi(1)}} \cdots V^{\operatorname{Set}(\beta)_{\chi(p)}} \cdot w\right). \tag{5.8}$$

However, if $\chi \in S_p$ satisfies $\beta_{\chi(i)} = \beta_i$ for each i, then

$$[\underline{w}]\left(V^{\operatorname{Set}(\beta)_{\chi(1)}} \cdots V^{\operatorname{Set}(\beta)_{\chi(p)}} w\right) = \begin{cases} 1, & \text{if } \chi = \mathrm{id}; \\ 0, & \text{if } \chi \neq \mathrm{id}, \end{cases} \tag{5.9}$$

because Lemma 6 (applied to $u = w$ and $\gamma = \beta$) shows that a term $\underline{v}$ in the expansion of the product $V^{\operatorname{Set}(\beta)_{\chi(1)}} \cdots V^{\operatorname{Set}(\beta)_{\chi(p)}}$ (where $v \in S_n$; keep in mind that this product belongs to the image of $\mathbf{k}[S_n]$ under $a \mapsto \underline{a}$) can satisfy $vw = w$ (that is, $\underline{v}\underline{w} = \underline{w}$) only if $\chi = \mathrm{id}$ and $v = \mathrm{id}$,

and it is easy to see (using Proposition 3) that the coefficient of $\underline{\mathrm{id}}$ in $V^{\mathrm{Set}(\beta)_{\chi(1)}} \cdots V^{\mathrm{Set}(\beta)_{\chi(p)}}$ is 1 for $\chi = \mathrm{id}$. Using (5.9), we can rewrite (5.8) as

$$[\underline{w}]\,(\mathbf{V}_\beta \mathbf{B}_\beta w) = \sum_{\substack{\chi \in S_p; \\ \beta_{\chi(i)} = \beta_i \text{ for each } i}} \begin{cases} 1, & \text{if } \chi = \mathrm{id}; \\ 0, & \text{if } \chi \neq \mathrm{id} \end{cases} = 1.$$

This proves part **(a)**.

(b) Assume that $u \neq w$, but we don't have $\widetilde{\beta} \prec_\pi \widetilde{\gamma}$. We must show that $[\underline{w}]\,(\mathbf{V}_\beta \mathbf{B}_\gamma u) = 0$. If we don't have $\widetilde{\beta} \preceq_\pi \widetilde{\gamma}$, then this is immediate from (5.1). So we WLOG assume that we do have $\widetilde{\beta} \preceq_\pi \widetilde{\gamma}$. Hence, $\widetilde{\beta} = \widetilde{\gamma}$ (since we don't have $\widetilde{\beta} \prec_\pi \widetilde{\gamma}$). Thus, the compositions β and γ are anagrams of one another, and in particular, have the same length; write them as $\beta = (\beta_1, \beta_2, \ldots, \beta_p)$ and $\gamma = (\gamma_1, \gamma_2, \ldots, \gamma_p)$. Now, Lemma 5 yields

$$\begin{aligned} [\underline{w}]\,(\mathbf{V}_\beta \mathbf{B}_\gamma u) &= [\underline{w}] \left(\sum_{\substack{\chi \in S_p; \\ \beta_{\chi(i)} = \gamma_i \text{ for each } i}} V^{\mathrm{Set}(\beta)_{\chi(1)}} \cdots V^{\mathrm{Set}(\beta)_{\chi(p)}} \cdot u \right) \\ &= \sum_{\substack{\chi \in S_p; \\ \beta_{\chi(i)} = \gamma_i \text{ for each } i}} [\underline{w}] \left(V^{\mathrm{Set}(\beta)_{\chi(1)}} \cdots V^{\mathrm{Set}(\beta)_{\chi(p)}} \cdot u \right). \end{aligned} \tag{5.10}$$

But each addend $[\underline{w}] \left(V^{\mathrm{Set}(\beta)_{\chi(1)}} \cdots V^{\mathrm{Set}(\beta)_{\chi(p)}} \cdot u \right)$ in this sum is 0, since Lemma 6 shows that a term $\underline{v}$ in the expansion of $V^{\mathrm{Set}(\beta)_{\chi(1)}} \cdots V^{\mathrm{Set}(\beta)_{\chi(p)}}$ can satisfy $vu = w$ (that is, $\underline{v}u = \underline{w}$) only if $\chi = \mathrm{id}$ and $v = \mathrm{id}$, but this is impossible (since $v = \mathrm{id}$ would simplify $vu = w$ to $u = w$, contradicting our assumption $u \neq w$). Hence, the whole sum in (5.10) is 0, and we obtain $[\underline{w}]\,(\mathbf{V}_\beta \mathbf{B}_\gamma u) = 0$. This proves part **(b)**. □

Proof of Theorem 2. We showed that $\mathcal{S}_\alpha \subseteq \mathcal{R}_\alpha$ in Section 3.3. Thus, it remains to prove that $\mathcal{R}_\alpha \subseteq \mathcal{S}_\alpha$. So let $b \in \mathcal{R}_\alpha$ be arbitrary. We must show that $b \in \mathcal{S}_\alpha$. The family $\{\mathbf{B}_{\mathrm{cLRM}'(w)}\, w\}_{w \in S_n} = \{\mathbf{B}_{\mathrm{LRM}'(w)}\, w\}_{w \in S_n}$ is a basis of $\mathcal{A}$ by Corollary 2. Hence, we can expand b as

$$b = \sum_{w \in S_n} \lambda_w \, \mathbf{B}_{\mathrm{cLRM}'(w)}\, w \tag{5.11}$$

with $\lambda_w \in \mathbf{k}$. We claim that $\lambda_w = 0$ for all $w \in S_n$ satisfying $\widetilde{\mathrm{cLRM}'(w)} \not\preceq_\pi \widetilde{\alpha}$. Once this claim is proved, (5.11) will reduce to a sum over the w that satisfy $\widetilde{\mathrm{cLRM}'(w)} \preceq_\pi \widetilde{\alpha}$, thus showing that $b \in \mathcal{S}_\alpha$ and thus concluding our proof. So we must only show our claim that $\lambda_w = 0$ for all $w \in S_n$ satisfying $\widetilde{\mathrm{cLRM}'(w)} \not\preceq_\pi \widetilde{\alpha}$. We prove this by strong induction on $\widetilde{\mathrm{cLRM}'(w)}$ (with respect to the reversed partition refinement order). That is, we fix a partition $\mu \vdash n$ with $\mu \not\preceq_\pi \widetilde{\alpha}$, and we assume as induction hypothesis that $\lambda_w = 0$ is already

proved for all $w \in S_n$ satisfying $\mu \prec_\pi \widetilde{\mathrm{cLRM}'(w)}$ (note that $\mu \prec_\pi \widetilde{\mathrm{cLRM}'(w)}$ and $\mu \not\preceq_\pi \widetilde{\alpha}$ automatically entail $\widetilde{\mathrm{cLRM}'(w)} \not\preceq_\pi \widetilde{\alpha}$, so we need not additionally require the latter). We then must prove that $\lambda_w = 0$ for all $w \in S_n$ satisfying $\widetilde{\mathrm{cLRM}'(w)} = \mu$.

Let $w \in S_n$ satisfy $\widetilde{\mathrm{cLRM}'(w)} = \mu$. Set $\beta := \mathrm{cLRM}'(w)$, so that $\widetilde{\beta} = \widetilde{\mathrm{cLRM}'(w)} = \mu \not\preceq_\pi \widetilde{\alpha}$. Hence, $\mathbf{V}_\beta \mathbf{B}_\alpha = 0$ by (5.1). Thus, $\mathbf{V}_\beta b = 0$ as well (because $b \in \mathcal{R}_\alpha = \mathbf{B}_\alpha \mathcal{A}$). Hence,

$$\begin{aligned} 0 = \mathbf{V}_\beta b &= \mathbf{V}_\beta \cdot \sum_{u \in S_n} \lambda_u \, \mathbf{B}_{\mathrm{cLRM}'(u)} \, u \qquad \text{(by (5.11))} \\ &= \sum_{u \in S_n} \lambda_u \, \mathbf{V}_\beta \mathbf{B}_{\mathrm{cLRM}'(u)} \, u. \end{aligned}$$

Taking the $\underline{w}$-coefficient, we thus conclude that

$$0 = [\underline{w}] \left(\sum_{u \in S_n} \lambda_u \, \mathbf{V}_\beta \mathbf{B}_{\mathrm{cLRM}'(u)} \, u \right) = \sum_{u \in S_n} \lambda_u \, [\underline{w}] \left(\mathbf{V}_\beta \mathbf{B}_{\mathrm{cLRM}'(u)} \, u \right). \tag{5.12}$$

But the sum on the right hand side here has many vanishing addends:

1. All the addends with $\mu \prec_\pi \widetilde{\mathrm{cLRM}'(u)}$ are 0 (since the induction hypothesis says that $\lambda_u = 0$ for all such u).

2. All the addends that satisfy $u \neq w$ but not $\mu \prec_\pi \widetilde{\mathrm{cLRM}'(u)}$ are 0 (since Lemma 7 **(b)**, applied to $\gamma = \mathrm{cLRM}'(u)$, yields $[\underline{w}] \left(\mathbf{V}_\beta \mathbf{B}_{\mathrm{cLRM}'(u)} \, u \right) = 0$).

The only remaining addend is the one for $u = w$; this addend equals λ_w (since we have $[\underline{w}] \left(\mathbf{V}_\beta \mathbf{B}_{\mathrm{cLRM}'(w)} \, w \right) = [\underline{w}] \left(\mathbf{V}_\beta \mathbf{B}_\beta w \right) = 1$ by Lemma 7 **(a)**). Thus, the whole sum simplifies to λ_w. Consequently, (5.12) rewrites as $0 = \lambda_w$, thus $\lambda_w = 0$, as desired. This completes the induction step, hence establishing our claim. □

6 Eigenvalues

We shall now elaborate upon and prove Corollary 3. We begin with notations for left and right multiplications on $\mathcal{A}$:

Definition 7. *Let $\mathbf{a} \in \mathcal{A}$. Then, we let $L(\mathbf{a}) : \mathcal{A} \to \mathcal{A}$ and $R(\mathbf{a}) : \mathcal{A} \to \mathcal{A}$ be the maps that send each $\mathbf{x} \in \mathcal{A}$ to $\mathbf{ax}$ and to $\mathbf{xa}$, respectively. These maps $L(\mathbf{a})$ and $R(\mathbf{a})$ are $\mathbf{k}$-linear, and are known respectively as* left *and* right multiplication *by $\mathbf{a}$.*

As linear endomorphisms of the free $\mathbf{k}$-module $\mathcal{A} = \mathbf{k}[S_n]$, these maps $L(\mathbf{a})$ and $R(\mathbf{a})$ can be represented by square matrices with respect to any basis of $\mathcal{A}$, and these matrices have eigenvalues, eigenvectors and characteristic polynomials. In general, their eigenvalues can fail to belong to $\mathbf{k}$; but not so when $\mathbf{a}$ belongs to the descent algebra Σ_n. Namely:

Corollary 4. *The* LRM-basis *shall mean the basis* $\left\{\mathbf{B}_{\mathrm{LRM}'(w)}\, w \;\middle|\; w \in S_n\right\}$ *of* $\mathcal{A}$ *from Corollary 2. We shall equip its indexing set* S_n *with a (strict) total order* $\lhd$*, in which two permutations* $u, v \in S_n$ *satisfy* $u \lhd v$ *if* $\mathrm{cLRM}'(u) \prec_\pi \mathrm{cLRM}'(v)$*. (This is, per se, only a partial order, but we pick any linear extension.)*

Let $\mathbf{a} = \sum_{\alpha \in \mathrm{Comp}_n} \lambda_\alpha \mathbf{B}_\alpha$ *(with* $\lambda_\alpha \in \mathbf{k}$*) be an arbitrary element of the descent algebra* Σ_n*. Then:*

(a) *The map* $L(\mathbf{a})$ *is represented by an upper-triangular matrix with respect to the LRM-basis.*

(b) *The eigenvalues of the map* $L(\mathbf{a})$ *(listed with their algebraic multiplicities) are the numbers* $\sum_{\alpha \in \mathrm{Comp}_n} \lambda_\alpha \eta_{\mathrm{cLRM}'(w)}(\alpha)$*, where* w *ranges over* S_n*, and where we use the* $\eta_\beta(\alpha)$ *defined in Theorem 1.*

Proof. **(a)** The map $L(\mathbf{a})$ depends $\mathbf{k}$-linearly on $\mathbf{a}$. Thus, from $\mathbf{a} = \sum_{\alpha \in \mathrm{Comp}_n} \lambda_\alpha \mathbf{B}_\alpha$, we obtain $L(\mathbf{a}) = \sum_{\alpha \in \mathrm{Comp}_n} \lambda_\alpha L(\mathbf{B}_\alpha)$. Hence, it suffices to show that each of the $L(\mathbf{B}_\alpha)$ is represented by an upper-triangular matrix with respect to the LRM-basis. In other words, it suffices to show that for each $\alpha \in \mathrm{Comp}_n$ and $v \in S_n$, we have

$$\begin{aligned} &\left(L(\mathbf{B}_\alpha)\right)\left(\mathbf{B}_{\mathrm{LRM}'(v)}\, v\right) \\ &= \left(\text{a linear combination of } \mathbf{B}_{\mathrm{LRM}'(u)}\, u \text{ with } u \trianglelefteq v\right) \end{aligned} \tag{6.1}$$

(where "$\trianglelefteq$" means "$\lhd$ or $=$"). Let $\alpha \in \mathrm{Comp}_n$ and $v \in S_n$. Set $\beta = \mathrm{cLRM}'(v)$. Then, $\mathbf{B}_\beta = \mathbf{B}_{\mathrm{LRM}'(v)}$. Meanwhile,

$$\begin{aligned} \left(L(\mathbf{B}_\alpha)\right)\left(\mathbf{B}_\beta\, v\right) &= \mathbf{B}_\alpha\ \mathbf{B}_\beta\, v \qquad \text{(by the definition of } L(\mathbf{B}_\alpha)\text{)} \\ &= \left(\eta_\beta(\alpha)\ \mathbf{B}_\beta + (\text{a sum of } \mathbf{B}_{\alpha'} \text{ with } \alpha' \prec \beta)\right) v \qquad \text{(by (2.3))} \\ &= \eta_\beta(\alpha)\ \mathbf{B}_\beta v + (\text{a sum of } \mathbf{B}_{\alpha'} v \text{ with } \alpha' \prec \beta)\,. \end{aligned}$$

In view of

$$\begin{aligned}
&(\text{a sum of } \mathbf{B}_{\alpha'} v \text{ with } \alpha' \prec \beta) \\
&\in \sum_{\substack{\alpha' \in \mathrm{Comp}_n; \\ \alpha' \prec \beta}} \mathcal{R}_{\alpha'} \qquad (\text{since } \mathbf{B}_{\alpha'} v \in \mathbf{B}_{\alpha'} \mathcal{A} = \mathcal{R}_{\alpha'}) \\
&\subseteq \sum_{\substack{\alpha' \in \mathrm{Comp}_n; \\ \widetilde{\alpha'} \prec_\pi \widetilde{\beta}}} \mathcal{R}_{\alpha'} \qquad \left(\text{because } \alpha' \prec \beta \text{ entails } \widetilde{\alpha'} \prec_\pi \widetilde{\beta}\right) \\
&= \sum_{\substack{\alpha' \in \mathrm{Comp}_n; \\ \widetilde{\alpha'} \prec_\pi \widetilde{\beta}}} \operatorname{span}\left\{ \mathbf{B}_{\mathrm{LRM}'(u)}\, u \;\mid\; \widetilde{\mathrm{cLRM}'(u)} \preceq_\pi \widetilde{\alpha'} \right\} \\
&\qquad \left(\text{by Theorem 2, summed over all } \alpha' \text{ with } \widetilde{\alpha'} \prec_\pi \widetilde{\beta}\right) \\
&= \operatorname{span}\left\{ \mathbf{B}_{\mathrm{LRM}'(u)}\, u \;\mid\; \widetilde{\mathrm{cLRM}'(u)} \prec_\pi \widetilde{\beta} \right\} \\
&\qquad \left(\text{since } \widetilde{\mathrm{cLRM}'(u)} \prec_\pi \widetilde{\beta} \text{ if and only if } \widetilde{\mathrm{cLRM}'(u)} \preceq_\pi \widetilde{\alpha'} \text{ for some } \widetilde{\alpha'} \prec_\pi \widetilde{\beta}\right) \\
&\subseteq \operatorname{span}\left\{ \mathbf{B}_{\mathrm{LRM}'(u)}\, u \;\mid\; u \lhd v \right\} \\
&\qquad \left(\text{since } \widetilde{\mathrm{cLRM}'(u)} \prec_\pi \widetilde{\beta} = \widetilde{\mathrm{cLRM}'(v)} \text{ entails } u \lhd v\right),
\end{aligned}$$

this leads to

$$\left(L\left(\mathbf{B}_\alpha\right)\right)\left(\mathbf{B}_\beta\, v\right) \in \eta_\beta\left(\alpha\right)\ \mathbf{B}_\beta\ v + \operatorname{span}\left\{ \mathbf{B}_{\mathrm{LRM}'(u)}\, u \;\mid\; u \lhd v \right\}.$$

In view of $\mathbf{B}_\beta = \mathbf{B}_{\mathrm{LRM}'(v)}$ and $\beta = \mathrm{cLRM}'\left(v\right)$, this rewrites as

$$\begin{aligned}
&\left(L\left(\mathbf{B}_\alpha\right)\right)\left(\mathbf{B}_{\mathrm{LRM}'(v)}\, v\right) \\
&\in \eta_{\mathrm{cLRM}'(v)}\left(\alpha\right)\ \mathbf{B}_{\mathrm{LRM}'(v)} v + \operatorname{span}\left\{ \mathbf{B}_{\mathrm{LRM}'(u)}\, u \;\mid\; u \lhd v \right\} \\
&\subseteq \operatorname{span}\left\{ \mathbf{B}_{\mathrm{LRM}'(u)}\, u \;\mid\; u \trianglelefteq v \right\},
\end{aligned} \tag{6.2}$$

which is precisely (6.1). Thus, Corollary 4 **(a)** is proved.

(b) By part **(a)**, the map $L\left(\mathbf{a}\right)$ is represented by an upper-triangular matrix with respect to the LRM-basis. Hence, the eigenvalues of $L\left(\mathbf{a}\right)$ are the diagonal entries of this matrix. But $L\left(\mathbf{a}\right) = \sum_{\alpha \in \mathrm{Comp}_n} \lambda_\alpha L\left(\mathbf{B}_\alpha\right)$ (as we saw in the proof of part **(a)**). Hence, for each $v \in S_n$,

we have

$$\begin{aligned}
&\left(L\left(\mathbf{a}\right)\right)\left(\mathbf{B}_{\mathrm{LRM}'(v)}\, v\right) \\
&= \sum_{\alpha \in \mathrm{Comp}_n} \lambda_\alpha \underbrace{\left(L\left(\mathbf{B}_\alpha\right)\right)\left(\mathbf{B}_{\mathrm{LRM}'(v)}\, v\right)}_{\substack{\in \eta_{\mathrm{cLRM}'(v)}(\alpha)\ \mathbf{B}_{\mathrm{LRM}'(v)} v + \mathrm{span}\left\{\mathbf{B}_{\mathrm{LRM}'(u)}\, u \mid u \lhd v\right\} \\ \text{(by (6.2))}}} \\
&\in \sum_{\alpha \in \mathrm{Comp}_n} \lambda_\alpha \left(\eta_{\mathrm{cLRM}'(v)}\left(\alpha\right)\ \mathbf{B}_{\mathrm{LRM}'(v)} v + \mathrm{span}\left\{\mathbf{B}_{\mathrm{LRM}'(u)}\, u \ \mid\ u \lhd v\right\}\right) \\
&\subseteq \left(\sum_{\alpha \in \mathrm{Comp}_n} \lambda_\alpha \eta_{\mathrm{cLRM}'(v)}\left(\alpha\right)\right) \mathbf{B}_{\mathrm{LRM}'(v)} v + \mathrm{span}\left\{\mathbf{B}_{\mathrm{LRM}'(u)}\, u \ \mid\ u \lhd v\right\}.
\end{aligned}$$

In other words, the diagonal entries of the matrix that represents $L\left(\mathbf{a}\right)$ with respect to the LRM-basis are the scalars $\sum_{\alpha \in \mathrm{Comp}_n} \lambda_\alpha \eta_{\mathrm{cLRM}'(v)}\left(\alpha\right)$ for all $v \in S_n$. So these scalars are the eigenvalues of $L\left(\mathbf{a}\right)$. This proves Corollary 4 **(b)**. □

Part **(b)** (but not part **(a)**) of Corollary 4 holds also for $R\left(\mathbf{a}\right)$ instead of $L\left(\mathbf{a}\right)$. This follows from Corollary 4 **(b)** via the following fact that has nothing to do with Σ_n:

Proposition 4. *Let $\mathbf{a} \in \mathcal{A}$ be arbitrary. Then:*

(a) *The linear maps $L\left(\mathbf{a}\right)$ and $R\left(\mathbf{a}\right)$ have the same characteristic polynomial, thus the same eigenvalues (with the same algebraic multiplicities).*

(b) *Let X be the matrix representing the map $L\left(\mathbf{a}\right)$ with respect to the basis $\left(w\right)_{w \in S_n}$ of $\mathcal{A}$. Let Y be the matrix representing the map $R\left(\mathbf{a}\right)$ with respect to the basis $\left(w^{-1}\right)_{w \in S_n}$ of $\mathcal{A}$. Then, $X = Y^T$.*

Proof. **(b)** By linearity, it suffices to prove this when $\mathbf{a}$ is a single permutation $p \in S_n$. In this case, the claim boils down to "$pu = v$ if and only if $v^{-1}p = u^{-1}$". But the latter is a trivial calculation.

(a) This follows from part **(b)**, since mutually transpose matrices have the same characteristic polynomial. □

7 Questions

Having proved the main results, we propose two questions.

1. On the basis $\left\{\mathbf{B}_{\mathrm{LRM}'(w)}\, w \ \middle|\ w \in S_n\right\}$ from Corollary 2, left multiplication by any element of Σ_n acts as a triangular matrix. However, right multiplication generally does not. Is there a basis that represents both left and right multiplication by triangular

matrices? Theoretical reasoning using the structure of Σ_n shows that the answer is "yes"[2], but does not suggest a concrete basis.[3]

2. A descent algebra can be defined for any finite Coxeter group [8]. It is desirable to know if an analogue of LRMs, and a basis such as the one in Corollary 2, still exists in this generality. Note that [2, §3] defines an analogue of LRMs in type B.

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[2]Indeed, if $\mathbf{k}$ is a field of characteristic 0, then the $\mathbf{k}$-algebra Σ_n becomes a direct product of $\mathbf{k}$'s after we quotient it by its Jacobson radical ([7, §5.2]). Thus, the same is true of the algebra $\Sigma_n \otimes \Sigma_n^{\mathrm{op}}$. Hence, each left $\Sigma_n \otimes \Sigma_n^{\mathrm{op}}$-module over a field of characteristic 0 has a basis on which $\Sigma_n \otimes \Sigma_n^{\mathrm{op}}$ acts by triangular matrices. In other words, each (Σ_n, Σ_n)-bimodule over a field of characteristic 0 has a basis on which Σ_n acts by triangular matrices from both left and right. This applies in particular to the (Σ_n, Σ_n)-bimodule $\mathcal{A}$. Using the structure of finitely generated modules over a PID, this can be extended to $\mathbf{k} = \mathbb{Z}$, and thus to all $\mathbf{k}$.

[3]NB: The analogue of Lemma 1 for the left ideals $\mathcal{L}_\alpha := \mathcal{A}\mathbf{B}_\alpha$ instead of the right ideals $\mathcal{R}_\alpha$ is false.

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